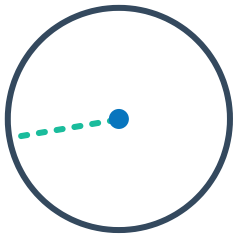
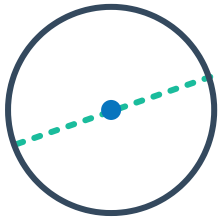
Features of the circle

1 Name the indicated part of the following circles:

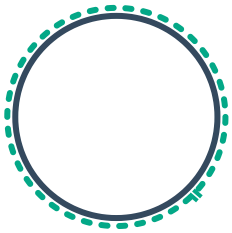
a



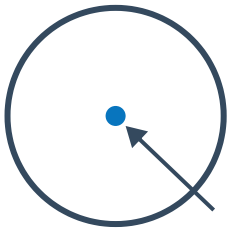
b



c

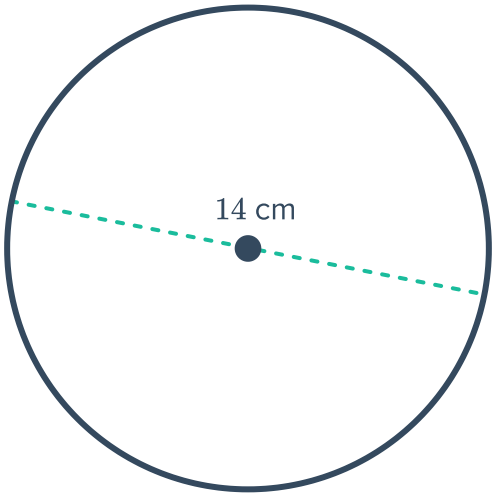


d

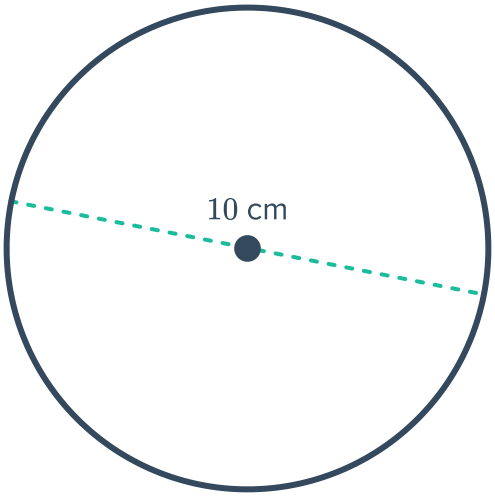


2 Find the radius of the following circles:

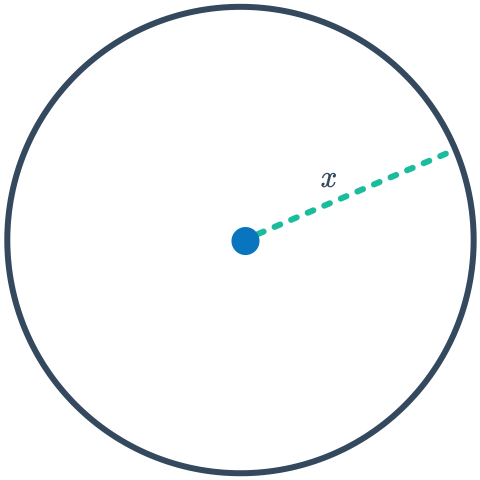
a



b



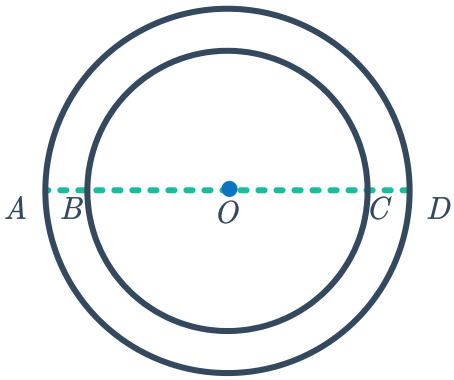
3 Find diameter of the following circle:



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4 The radius of the smaller circle shown in the figure is 16 cm:

If $CD = 4$ cm, find AD, the diameter of the larger circle.



5 Complete the following tables:

a

	Diameter (cm)	Radius (cm)
Circle 1		11
Circle 2		17
Circle 3		19
Circle 4		21
Circle 5		23

b

	Diameter (cm)	Radius (cm)
Circle 1	22	
Circle 2	26	
Circle 3	28	
Circle 4	36	
Circle 5	42	

Circumference

6 Determine whether the following statements are true about π.

a π is equal to 3.142.

b π cannot be expressed as a fraction using whole numbers.

c π is the ratio between the circumference and the diameter of a circle.

d π is equal to $\frac{22}{7}$.

e π is the ratio between the circumference and the radius of a circle.

f π is equal to 3.14.

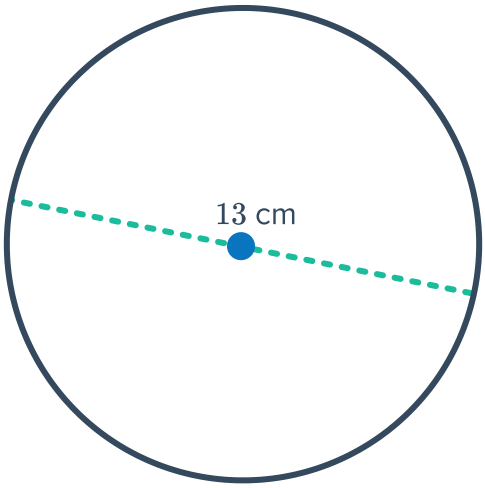
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7 For each of the following circle:

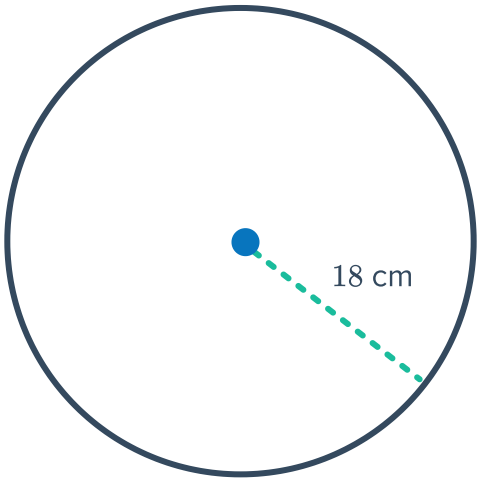
i Calculate the exact circumference in terms of π.

ii Hence, calculate the circumference rounded to two decimal places.

a



b



c A circle with a diameter of $7.7\,\text{cm}$.

d A circle with a radius of $4.1\,\text{cm}$.

8 Calculate the exact circumference of the following circles:

a A circle with a diameter of $59\,\text{cm}$.

b A circle with a radius of $22\,\text{cm}$.

9 Calculate the circumference of the following circles, correct to two two decimal places:

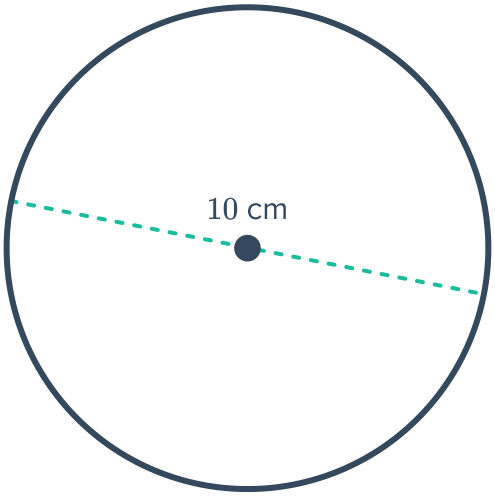
a A circle with a diameter of $38\ \text{cm}$.

b A circle with a diameter of $45\ \text{cm}$.

c A circle with a radius of $27\ \text{cm}$.

d A circle with a radius of $1.8\ \text{cm}$.

10 Caitlin and David calculate the circumference of this circle using different formulas.



a Caitlin uses the formula $C = \pi d$ to calculate the circumference.

Find her result as an exact value

b David uses the formula $C = 2\pi r$ to calculate the circumference, where r is the radius.

Find his result as an exact value.

c Compare the results of their calculations and explain what you notice.

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11 A circle has a circumference of $52\,\text{cm}$.

- **a** Calculate the exact radius of the circle.
- **b** Hence, calculate the radius rounded to two decimal place.

12 Calculate the exact radius of the following circles:

- **a** A circle with circumference of $12\pi\,\text{cm}$.
- **b** A circle with circumference of $12\,\text{cm}$.
- **c** A circle with circumference of $32\pi\,\text{cm}$
- **d** A circle with circumference of $26\,\text{cm}$

13 Calculate the radius of the following circles, correct to two decimal places:

- **a** A circle with circumference of $14\,\text{cm}$.
- **b** A circle with circumference of $18\,\text{cm}$.
- **c** A circle with circumference of $14.9\,\text{cm}$.
- **d** A circle with circumference of $98\,\text{cm}$.

14 A circle has a circumference of $36\,\text{cm}$.

- **a** Calculate the exact diameter of the circle.
- **b** Hence, calculate the diameter rounded to two decimal place.

15 Calculate the exact diameter of the following circles:

- **a** A circle with circumference of $18\pi\,\text{cm}$.
- **b** A circle with circumference $18\,\text{cm}$.
- **c** A circle with circumference of $17\pi\,\text{cm}$.
- **d** A circle with circumference of $14\,\text{cm}$.

16 Calculate the diameter of the following circles, correct to two decimal places:

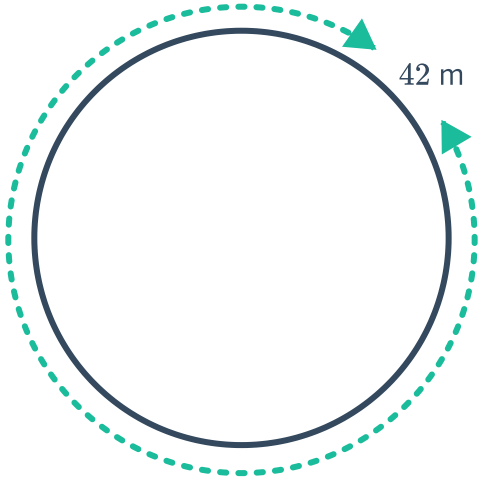
- **a** A circle with circumference of $37\,\text{cm}$.
- **b** A circle with circumference of $44\,\text{cm}$.
- **c** A circle with circumference of $70\,\text{cm}$.
- **d** A circle with circumference of $24.5\,\text{cm}$.

17 Complete the following table that lists the exact radii, diameters, and circumferences of various circles:

	Radius (cm)	Diameter (cm)	Circumference (cm)
Circle 1	2	4	
Circle 2	8		
Circle 3		82	
Circle 4			340

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18 Consider the given circle:



a Find the diameter D, correct to two decimal places.

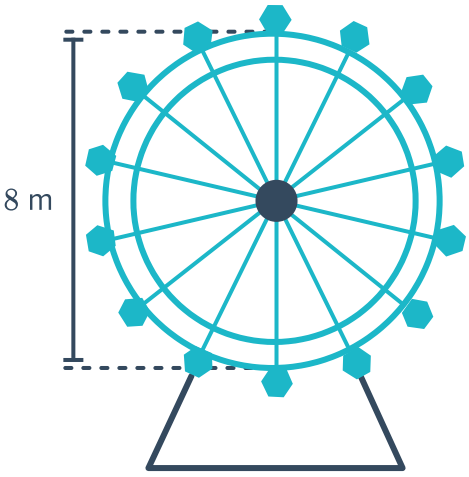
b Find the radius r, correct to two decimal places.

Applications

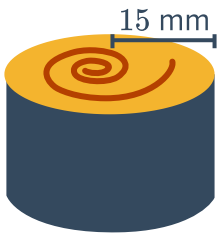
19 The bottom of a flower pot has a radius of $16\ \text{cm}$. Find the circumference of the bottom of the flower pot, correct to one decimal place.

20 A scooter tyre has a diameter of $34\ \text{cm}$. Find the circumference of the tyre, correct to one decimal place.

21 Find the circumference of the Ferris wheel, correct to one decimal place.



22 Find the length of the strip of seaweed around the outside of the sushi shown. Round your answer correct to one decimal place.



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23 A coin has a diameter of 2.2 cm. If the coin is rolled through 45 complete revolutions, find the distance it will travel. Round your answer to one decimal place.

24 A circular running track has a diameter of 23 m. How many laps must be completed to run 1600 m? Round your answer to one decimal place.

25 The diameter of the front wheel of a car is 0.4 m. How many complete revolutions will the wheel make if the car travels a distance of 50 km?

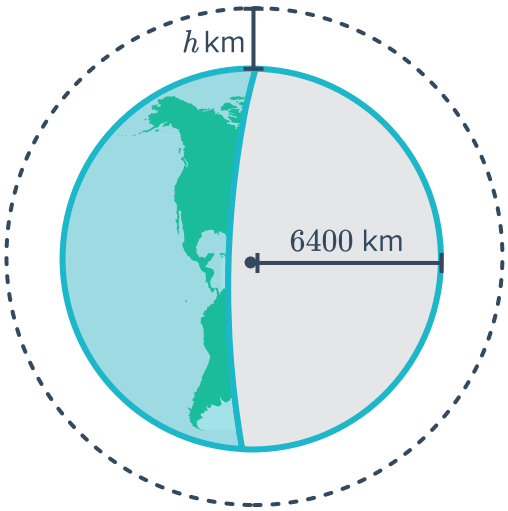
26 The wheel of Roxanne's bicycle has a radius of 22 cm. How far would she cycle if the wheels made 250 complete revolutions? Give your answer in metres, correct to one decimal place.

27 Carl is performing an experiment by spinning a metal weight around on the end of a nylon thread. How far does the metal weight travel if it completes 40 revolutions on the end of a 0.65 m thread? Round your answer to one decimal place.

28 A shallow circular wading pool has a diameter of 4 m. A deeper circular swimming pool has a radius of 16 m. How many times greater is the circumference of the swimming pool than the wading pool?

29 A record has a diameter of 30 cm and is played at a speed of $33\frac{1}{3}$ revolutions per minute. Through what total distance will a point on the rim of the record travel if the record takes 28 minutes to play? Give your answer in metres, correct to one decimal place.

30 A satellite is orbiting the Earth at a height of h km above the Earth's surface. In one complete orbit, the satellite travels a distance of $41\,664$ km. If the radius of the Earth is 6400 km, find the height of the satellite above the Earth. Round your answer to one decimal place.



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